



Large-Scale Sentiment and Topic Analysis of Reviews



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Project Overview

Objective :

The objective of this project is to build a machine learning model capable of classifying Yelp reviews into positive (1) or negative (0) based on the textual content of the reviews.

The project follows a complete machine learning pipeline including data loading, preprocessing ,class balancing, text normalization, feature extraction, model training, evaluation, visualization, and topic modeling.

Expected Outcomes :

By the end of this project, we expect to :

- **Train and compare two machine learning models:**
 - SGD Classifier
 - Random Forest
- **Evaluate model performance using:**
 - Accuracy
 - Precision
 - Recall
 - F1-score
 - Confusion Matrix
 - ROC Curve
- **Identify the best-performing model**
- **Extract important topics using LDA**

DATASET

Dataset Source :

The dataset used in this project is the Yelp Academic Dataset provided in JSON format.



Dataset Description :

The dataset consists of :

- A text column containing customer reviews
- A star rating column (1–5)

The dataset is unstructured textual data stored in JSON format, which requires preprocessing before applying machine learning models.

The sentiment label is defined as:

- (1) → Positive (stars ≥ 4)
- (0) → Negative (stars < 4)

Data Preparation :

Handling Missing Values

Missing values were checked and removed to ensure data quality.

Sentiment Labeling

Star ratings were converted into binary sentiment labels.

Class Balancing

To prevent model bias, undersampling was applied to balance the dataset.

The final balanced dataset size is :

121,372 records (60,686 per class)

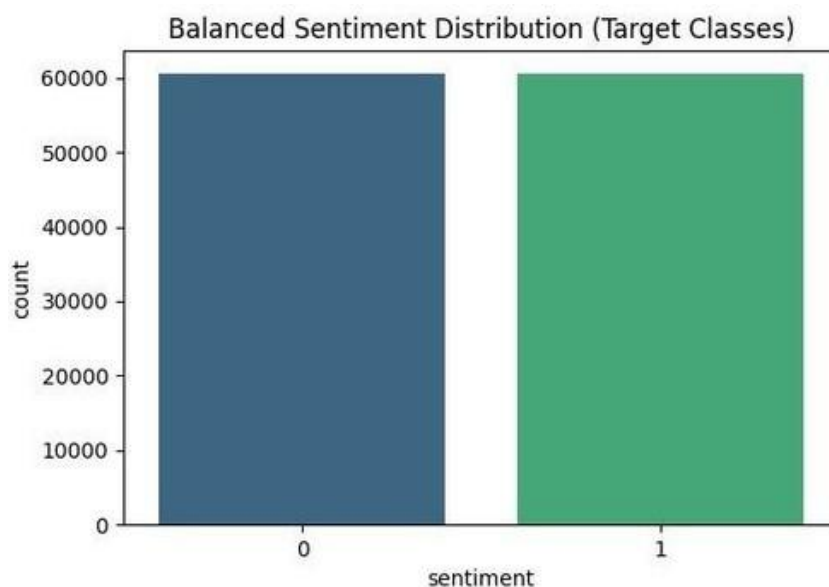


Figure 1: This figure shows that both sentiment classes are equally distributed after applying undersampling, ensuring fair and unbiased model training.

EXPLORATORY DATA ANALYSIS (EDA)

Word Cloud :

A word cloud was generated to visualize the most frequent words in positive reviews.



Figure 2: The word cloud highlights frequently used terms such as "good", "great", and "food", indicating key factors in positive customer experiences.

MODEL SELECTION AND TRAINING

Model Choice :

Two machine learning models were selected :

- SGD Classifier (efficient for high-dimensional text data)
- Random Forest (ensemble model for capturing complex patterns)

Feature Engineering :

TF-IDF (Term Frequency–Inverse Document Frequency) was used to convert text into numerical features using :

- ◆ Unigrams
- ◆ Bigrams

Train-Test Split :

The dataset was split into :

- 80% training
- 20% testing

MODEL EVALUATION AND TESTING

Evaluation Metrics :

The following metrics were used :

- Accuracy
- Precision
- Recall
- F1-score

Model Performance Comparison :

Model	Accuracy	Precision	Recall	F1-score
SGD	87.5%	0.88	0.87	0.88
Random Forest	81.8%	0.81	0.84	0.82

The SGD model outperforms Random Forest across all evaluation metrics, achieving higher accuracy and more balanced precision and recall. This indicates that SGD is more suitable for high-dimensional sparse text data such as TF-IDF representations.



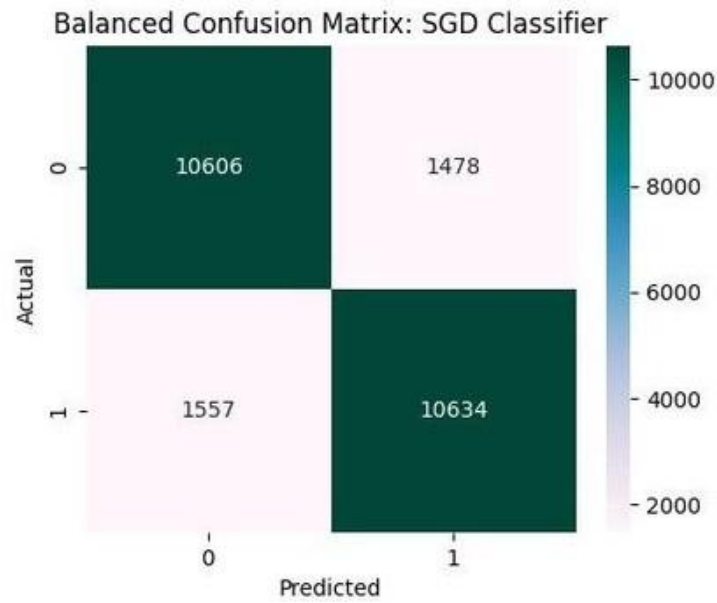


Figure 3 : The SGD model correctly classifies most reviews with fewer misclassification errors.

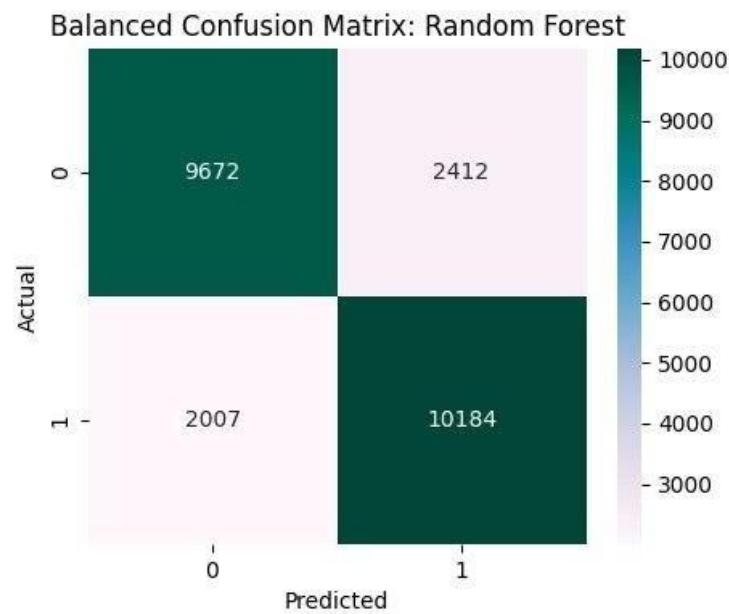


Figure 4 : The Random Forest model shows more classification errors, especially in distinguishing between classes.



ROC Curve and AUC :

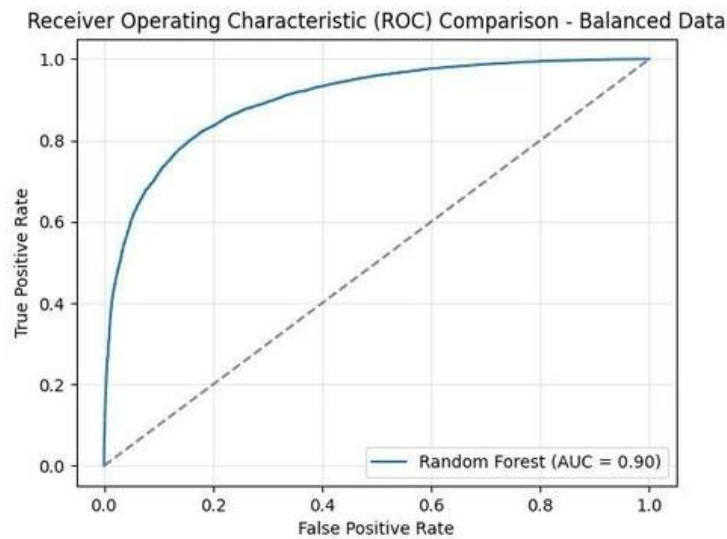


Figure 5 : The ROC curve illustrates the trade-off between the true positive rate and false positive rate.

The Random Forest model achieved an AUC of approximately 0.90.

The ROC curve comparison further supports that the SGD model performs better overall than Random Forest.

An AUC score of 0.90 indicates that the model has a strong ability to distinguish between positive and negative reviews.

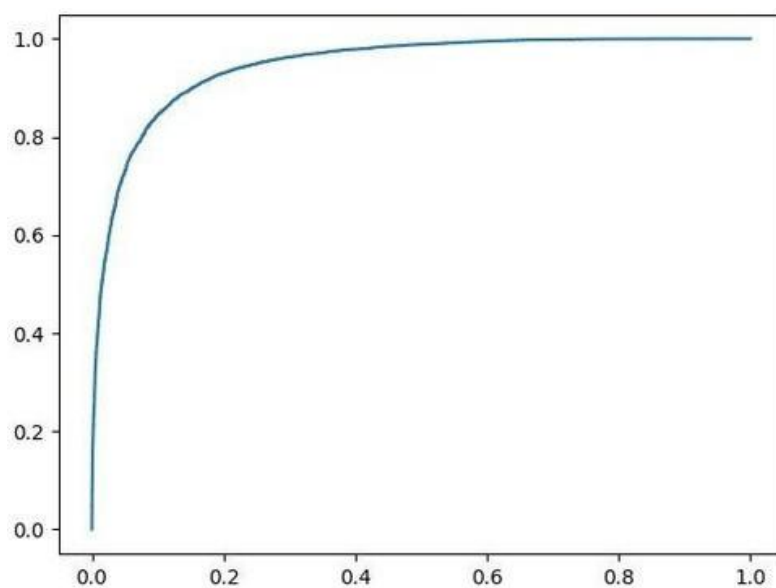


Figure 6 : This confirms the model's ability to distinguish between classes.

Result Analysis :

The results indicate that the SGD Classifier outperforms the Random Forest model in terms of accuracy and overall performance.

This is mainly because text data represented using TF-IDF is high-dimensional and sparse, which is better handled by linear models such as SGD. In contrast, Random Forest struggles with such data due to its tree-based structure.

Additionally, class balancing significantly improved model fairness, reducing bias toward the majority class.

These results confirm that linear models are more suitable for high-dimensional sparse text data.

DISCUSSION

The results clearly show the following:

- The SGD Classifier achieved better performance than Random Forest.
- Linear models perform better for high-dimensional sparse text data.
- Class balancing improved model performance and fairness.
- TF-IDF effectively captured important textual features.

However:

- Random Forest is less suitable for sparse text data.
- Some misclassification errors still exist.

Additionally:

- The SGD model achieved higher accuracy (87.5%) compared to Random Forest (81.8%), with more balanced precision, recall, and F1-score values.
- The confusion matrix shows that SGD produces fewer misclassification errors, indicating better prediction reliability.
- The ROC curve analysis also supports this result, demonstrating that SGD has a stronger ability to distinguish between positive and negative reviews.

Overall :

- These findings confirm that choosing the right model based on data characteristics is critical.
- Linear models are more suitable for sentiment analysis tasks involving high-dimensional sparse text data

CONCLUSION

In this project, a complete machine learning pipeline was implemented for sentiment analysis of Yelp reviews.

Key Achievements :

- Processing a large real-world dataset
- Applying text preprocessing techniques
- Training and comparing two machine learning models
- Evaluating performance using multiple metrics
- Using visualization tools such as confusion matrices and ROC curves

The SGD Classifier achieved the best performance and was selected as the final model.

Challenges :

- Handling a large dataset efficiently
- Cleaning and preprocessing unstructured text data
- Managing class imbalance in the dataset
- Working with high-dimensional feature space

Lessons Learned :

- The importance of data preprocessing in machine learning
- The impact of class imbalance on model performance
- The effectiveness of TF-IDF in text representation
- The advantage of linear models for text classification tasks



```
Project ▾ absa_trend_analyzer.py x yelp_academic_dataset_review.json
Run absa_trend_analyzer x
C:\Users\shado\PycharmProjects\PythonProject19\venv\Scripts\python.exe C:\Users\shado\PycharmProjects\PythonProject19\absa_trend_analyzer.py
Step 1: Ingesting Dataset and Auditing Data Quality...

[Data Integrity Audit - Initial Missing Values]
text      0
stars     0
dtype: int64

Applying Class Balancing Strategy (Undersampling)...
Final Balanced Dataset Size: 121372 records (60686 per class).

Step 2: Executing Text Normalization Pipeline...
Step 3: Generating Exploratory Visualizations...
Step 4: Vectorizing Text and Training Comparative Models...

Step 5: Performing Unsupervised Aspect Extraction (LDA)...
Aspect 1:
['good', 'fri', 'order', 'chicken', 'like', 'pizza', 'burger', 'food', 'sauc', 'place']
Aspect 2:
['room', 'get', 'time', 'hotel', 'store', 'go', 'work', 'car', 'one', 'mould']
Aspect 3:
['food', 'taco', 'good', 'place', 'mexican', 'go', 'price', 'mait', 'salsa', 'like']
Aspect 4:
['order', 'food', 'mait', 'us', 'time', 'minut', 'servic', 'tabl', 'ask', 'get']
Aspect 5:
['great', 'place', 'food', 'good', 'love', 'servic', 'friendli', 'beer', 'coffe', 'delici']

Step 6: Generating Performance Analytics...

Results for SGD Classifier:
Accuracy Index: 87.50%
precision  recall  f1-score  support
```